

- Document Control
- System Name: Application
- Version:
- Owner:
- Date:
- Status:
  
- Executive Summary
  - This BRD outlines the business requirements for processing source CSV customer and order data in the Application, transforming, cleaning, aggregating, and loading to multiple managed delta tables using delta live tables.
  
- Business Objectives
  - Streamline loading, cleaning, and processing of customer and order data.
  - Ensure data quality and deduplication across customer/order tables.
  - Enable data aggregation for customer spend analysis.
  - Maintain historical changes with SCD Type 2 methodology.
  
- In-Scope
  - Loading customer and order data from CSV to delta tables.
  - Schema management for customer and order data.
  - Calculation and addition of TotalAmount in order data.
  - Data cleaning (null and duplicate removal).
  - Ordersummary SCD Type 2 table creation and management.
  - Aggregation and storage of customer spend data.
  - End-to-end implementation using delta live tables.
  
- Out-of-Scope
  - Real-time data streaming integrations.
  - External system (non-delta) data processing.
  - Visualization or BI reporting.
  - Integration with non-catalog/sdlc\_wizard schemas.
  
- Stakeholders
  - Data Engineers:
  - Business Analysts:
  - IT/Data Governance:
  - End Users:

- Business Requirements

#### ### BR-APP-001 — Load Customer and Order Data

- Description: Read source CSV data and load into delta tables customer and order.
- Business Value: Centralized data management and governance.
- Priority: High

#### ### BR-APP-002 — Schema Definition for Customer and Order Tables

- Description: Ensure schema of customer (CustId, Name, EmailId, Region) and order (OrderId, ItemName, PricePerUnit, Qty, Date, CustId) is enforced.
- Business Value: Consistency and data quality.
- Priority: High

#### ### BR-APP-003 — Calculate TotalAmount in Orders

- Description: Add a TotalAmount column in order table ( $\text{PricePerUnit} \times \text{Qty}$ ).
- Business Value: Prepares data for downstream aggregation and reporting.
- Priority: High

#### ### BR-APP-004 — Data Cleaning for Null and Duplicate Records

- Description: Remove records with Null values and duplicates from both tables.
- Business Value: Maintains trust and accuracy in datasets.
- Priority: High

#### ### BR-APP-005 — Ordersummary Table Creation

- Description: Create ordersummary table with required schema if it does not exist.
- Business Value: Provides a single view of customer's order data.
- Priority: High

#### ### BR-APP-006 — Join Customer and Order Data for SCD2

- Description: Join customer and order by CustId and load data in SCD type 2 table ordersummary.
- Business Value: Enables historical tracking of data changes.
- Priority: High

#### ### BR-APP-007 — SCD Type 2 Update Logic

- Description: Implement update logic for SCD type 2 to maintain active/inactive records and history.
- Business Value: Preserves data audit trail.
- Priority: High

#### ### BR-APP-008 — Customer Aggregate Spend Table

- Description: Create and maintain customeraggregatespend table with Name, TotalAmount, and Date.
- Business Value: Supports business analysis for aggregated spend.
- Priority: Medium

#### ### BR-APP-009 — Aggregate TotalAmount by Name and Date

- Description: Aggregate TotalAmount by Name and Date from ordersummary into customeraggregatespend.
- Business Value: Facilitates analytical needs for customer spend.
- Priority: Medium

#### ### BR-APP-010 — End-to-End Implementation via Delta Live Tables

- Description: Implement steps 1 through 10 using delta live tables.
- Business Value: Automation, reliability, and scalability.
- Priority: High
- Assumptions & Constraints
  - Source data is available in expected CSV format.
  - Delta table environment is pre-configured.
  - Only specified schemas and tables are in scope.
  - Delta live tables environment is available.
- Success Metrics
  - 100% accuracy in schema enforcement and transformation.
  - No duplicate or null records post-processing.
  - Successful SCD2 logic with correct historical tracking.
  - Daily automated runs without failure.
- Risks & Mitigations
  - Source data quality issues: Establish automated quality checks.
  - Delta live tables performance constraints: Regular monitoring and optimization.
  - Incorrect SCD2 handling: Design validation and peer review.
- Approval
  - Project Sponsor:
  - Data Management Lead:
  - Application Owner: